Signals, Systems & Probability Homework 6

Notice: This assignment is primarily coding-based. You are expected to submit one PDF file and three .m source code files (one for each problem). In your PDF, include the answers to each question along with the generated plots and clear explanations.

Problem 1

Refer to the signal for $T = 3$ from In-class Problem #3 in Lecture 5. $x(t) = e^{-t} \quad 0 \le t < T$

(a) Draw $x(t)$ in MATLAB (show multiple periods).

(b) Find its Fourier series coefficients using the theoretical method (show a_k as a function of k).

(c) Find its Fourier series coefficients using the empirical method (using "mean" in MATLAB). Compare the results with (b).

(d) Use the result from (c) to reconstruct the signal and compare it with (a). Show the effect of N. What would be a sufficient N to have a good approximation of $x(t)$ in your simulation?

Problem 2

Refer to the signal for $T = 2$ from Problem 3 in HW5. $x(t) = \cos\frac{2\pi}{T}t + \left|\cos\frac{2\pi}{T}t\right|$

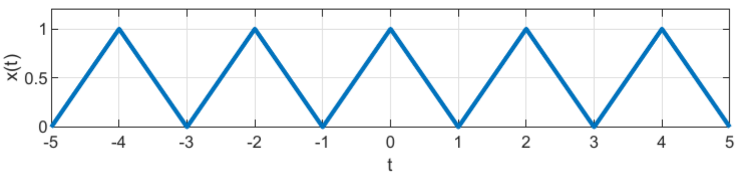
(a) Draw $x(t)$ in MATLAB (show multiple periods).

(b) Find its Fourier series coefficients using the empirical method. Compare them with your solutions in HW5.

(c) Use the result from (b) to reconstruct the signal and compare it with (a). Show the effect of N. What would be a sufficient N to have a good approximation of $x(t)$ in your simulation?

Problem 3

Refer to the following periodic signal with $T = 2$



(a) Find its Fourier series coefficients using the theoretical method.

(b) Find its Fourier series coefficients using the empirical method. Compare the results with (a).

(c) Use the result from (b) to reconstruct the signal and compare it with the original signal $x(t)$. Show the effect of N. What would be a sufficient N to have a good approximation of $x(t)$ in your simulation?